

MANISH MAN SHRESTHA

Firmware & Embedded Systems Engineer | IoT Developer

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Hiring Team

John Deere — Intelligent Solutions Group
Dubuque, Iowa

Re: Embedded Software Engineer, Displays (Requisition 120793)

Dear Hiring Team,

I am excited to apply for the Embedded Software Engineer, Displays role. Few openings align as closely with my background as this one: I have spent 15 years building real-time embedded firmware — including on-device display and HMI software — and my doctoral research centers on the very domain John Deere leads, technology that helps agriculture serve the world.

Here is how my experience maps to what the role calls for:

- **Embedded C/C++ product development:** I develop real-time firmware in Embedded C/C++ across ARM Cortex-M (STM32, NXP/LPC) and ESP32 — device drivers, board bring-up, and RTOS-based architectures delivered on real hardware.
- **Display / HMI software:** I designed and integrated a custom I2C SSD1306 OLED display driver for live device status and measurement feedback on a multichannel monitoring instrument, and built LCD and push-button interfaces for production panel meters.
- **Agricultural and industrial machine environments:** My Ph.D. in Agricultural & Biosystems Engineering (completing May 2026) involved designing and field-deploying embedded sensing and control systems for greenhouse and hydroponic production — including a 120-day field deployment that ran with zero failures. I understand ag machine environments from the sensor up.
- **Delivering to production with cross-functional teams:** As a firmware engineer I shipped firmware for four distinct commercial products, and as a CTO I led embedded systems from concept through hardware integration and production validation, using Git-based version control and CI/CD.
- **Linux and application software:** I develop on Linux-based platforms (Raspberry Pi) and have built supporting applications and dashboards (C#/.NET, ASP.NET) for real-time data retrieval and visualization.

What draws me specifically to John Deere is the mission fit. I have spent my research career making controlled-environment agriculture smarter and more reliable; bringing that perspective to the displays farmers rely on every day would be a natural and motivating next step. I am based in Brookings, SD and am glad to relocate to Dubuque.

I would welcome the chance to discuss how my embedded firmware and agricultural-systems background can contribute to your display software team. Thank you for your consideration.

Sincerely,

Manish Man Shrestha